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Pila


Octopus


Squid


Oyster

Figure 2.16 Examples of Mollusca

digestive system is complete and mouth contains a rasping organ called radula with transverse rows of chitinous teeth for feeding (radula is absent in bivalves. The sense organs are tentacles, eyes and ospharidium (to test the purity of water and present in bivalves and gastropods). Excretory organs are nephridia. Open type of circulatory system is seen except for cephalopods such as squids, cuttle fishes and octopuses. Blood contains haemocyanin, a copper containing respiratory pigment. They are dioecious and oviparous. Development is indirect with a veliger larva (a modified trochophore larva).

Examples: *Pila* (Apple snail), *Lamellidens* (Mussel), *Pinctada* (Pearl oyster), *Sepia* (Cuttle fish), *Loligo* (Squid), *Octopus* (Devil fish) (Figure 2.16).



Marbled Cone Snail
(*Conus marmoreus*)

This cone-shaped snail can deliver dangerous venom which may result in vision loss, respiratory failure, muscle paralysis and eventually death. There is no anti-venom available.



2.3.9 Phylum Echinodermata

(G. *Echinos* – spiny; *dermos* –skin)

All Echinoderms are marine animals. The adults are radially symmetrical but the larvae are bilaterally symmetrical. These animals have a mesodermal endoskeleton of calcareous ossicles and hence the name Echinodermata (spiny skin). They are exclusively marine with organ system level of organisation. The most distinctive feature of echinoderms is the presence of the water vascular system or ambulacral system with tube feet or podia, which helps in locomotion, capture and transport of food and respiration. The digestive system is complete with mouth on ventral side and anus on the dorsal side. Excretory organs are absent. The nervous system and sensory organs are poorly developed. The circulatory system is


Starfish


Brittle star


Sea cucumber


Sea urchin

Figure 2.17 Examples of Echinodermata